Networks and Complexity

Solution 21-10

This is an example solution from the forthcoming book Networks and Complexity. Find more exercises at https://github.com/NC-Book/NCB

Ex 21.10: Leaky conveyor [Lvl. 4]

Consider a random walker on a network described by a weighted, directed adjacency matrix

$$\mathbf{A} = \begin{pmatrix} 0 & 1 & 0 \\ 2 & 0 & 1 \\ 1 & 2 & 0 \end{pmatrix}$$

the element A_{ij} describes the rate at which the walker that is in node j jumps to node i. Determine the probability that the walker is in node 1 in the stationary state. [The probabilities become stationary, not the walker.]

Solution

The differential equations that describe this system are

$$\begin{aligned} \dot{x}_1 &= -3x_1 + x_2 && (1) \\ \dot{x}_2 &= -3x_2 + 2x_1 + x_3 && (2) \\ \dot{x}_3 &= -x_3 + x_1 + 2x_2 && (3) \end{aligned}$$

which we can write in matrix form

$$\dot{\boldsymbol{x}} = -\mathbf{L}\boldsymbol{x} \tag{4}$$

where

$$\mathbf{L} = \begin{pmatrix} 3 & -1 & 0 \\ -2 & 3 & -1 \\ -1 & -2 & 1 \end{pmatrix} \tag{5}$$

To find the long term behavior we need to find the eigenvector with eigenvalue 0. (There needs to be one. Otherwise, the walker number would not be conserved.) Since we already know that the eigenvalue we are looking for is zero, we can move straight to the conditions

$$\begin{aligned} 0 &= -3v_1 + v_2 && (6) \\ 0 &= -3v_2 + 2v_1 + v_3 && (7) \\ 0 &= -v_3 + v_1 + 2v_2 && (8) \end{aligned}$$

Of course, this is the same as finding the steady state of the dynamical system, just with v instead of x. The first equation tells us

$$v_2 = 3v_1 \tag{9}$$

which we can use to eliminate v_2,

$$\begin{aligned} 0 &= -9v_1 + 2v_1 + v_3 = -7v_1 + v_3 && (10) \\ 0 &= -v_3 + v_1 + 6v_1 = 7v_1 - v_3 && (11) \end{aligned}$$

Hence the eigenvector is

$$\boldsymbol{v_1} = \begin{pmatrix} 1 \\ 3 \\ 7 \end{pmatrix} \tag{12}$$

Since we are talking about probabilities, we want to normalize the vector such that it sums to one. So, after a long time the probability of finding the walker in node 1 is just 1/11.